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2.2 - Narrow-Complex Tachycardia 2: AFib, & AFlutter SOA, MAT SOAP

1(O) Elevated heart clock:
Supraventricular tachycardias (SVTs) typically present as narrow complex (QRS < 120ms, 3 boxes) **tachycardias (>100 bpm)** → A fib ~100-200 bpm; A flutter ~150 bpm w/ 2:1 block; AT ~140-220 bpm; MAT >100 bpm vs multifocal atrial rhythm/"wandering pacemaker" <100 bpm

2(O) "Narrow pine" less than three boxes:
AFib, AFlutter, MAT, AT typically present as **narrow complex (QRS < 120ms, 3 boxes)**tachycardias (>100 bpm) → impulse originates at/above AV node (supraventricular) to His Purkinje system of ventricles → aberrant conduction (bundle branch block, accessory pathway) may cause wide complex

3) Venous connections to TV:
Atrial fibrillation (AFib) is caused by continuous, chaotic electrical signals from all over the atria → ectopic foci in myocardial tissue surrounding **pulmonary veins (AFib/ Intro)**

4) Irregularly irregular static on screen:
AFib is caused by continuous, chaotic electrical signals from all over the atria (>350 signals/min) → some signals transmitted to ventricles through AV node (~100-200 bpm depending on refractory period of node, ~90-170 bpm in older pts) → **"irregularly irregular"** rhythm (different R-R intervals in no obvious pattern) (**AFib/ ECGM**)

5) Wavy baseline static:
no distinct P waves in the setting of AFib → wavy erratic baseline (**f waves**) between the transmitted QRS complexes (**AFib/ ECG**)

6) "Episodes" + "this week":
Paroxysmal AFib = intermittent or resolves within **7 days** of onset (spontaneously or with meds) (**AFib/ HPI**)

7) "Persistent" + "into next week":
AFib often progresses from paroxysmal (episodes, ≤7 days) to **persistent AFib (>7 days)**; permanent AFib = rhythm control strategy no longer pursued (**AFib/ HPI**)

8) Dog with "fluttering" ears running around tricuspid bowl:
"Typical" Atrial flutter (AFlutter) is caused by a **reentrant circuit around the tricuspid annulus**, through the cavotricuspid isthmus (vs "atypical" AFlutter caused by single area of scar tissue in atrium) → atria ~300 bpm ("sawtooth" flutter waves) (**AFlutter/ Intro**)

9(O) Dog with “fluttering” ears and regular intermittent jumps:
no distinct P waves in the setting of AFlutter → **“sawtooth” flutter waves (F waves)**~300 bpm → typically 1:2 signals filtered through AV node (1:2 block) → regular narrow QRS complexes (ventricular rate ~150 bpm) (**AFlutter/ ECG**)

10(O) “300” -note- “150” walker blocking dog jumps:
AFlutter → “sawtooth” flutter waves (F waves) ~300 bpm → typically 1:2 signals filtered through AV node **(1:2 block)** → **regular narrow QRS complexes (ventricular rate ~150 bpm)**; other ratios (e.g. 1:3, 1:4) also common **(AFlutter/ ECG)**

11(O) Dog in front if irregular notes jumping erratically:
AFlutter → “sawtooth” flutter waves (F waves) ~300 bpm → filtered through AV node to ventricles → typically no variability in AV nodal block = regular tachycardia; **variable AV nodal block = irregular tachycardia (different R-R intervals) (AFlutter/ ECG)**

12(S) Oblivious laughter:
AFib/AFlutter can be **asymptomatic (AFib/AFlutter/ HPI)**

13(S) Palpating chest:
AFib/AFlutter most commonly presents with **palpitations (AFib/AFlutter/ Chief Complaint)**

14(S) Failing heart balloon:
AFib/AFlutter causes loss of atrial kick (↓LV filling) + tachycardia/dysrhythmia hemodynamic changes + tachycardia-mediated cardiomyopathy (chronic) → ↓cardiac output → **acute/chronic heart failure (AFib/AFlutter/ Chief Complaint)**

15(S) Fainting + out of breath + dilated pants:
AFib/AFlutter can present with signs/symptoms of **acute/chronic heart failure** → hypotension (**dizziness/lightheadedness/AMS), periph edema**, JVD, pulmonary edema, cardiac gallop, **exertional dyspnea/fatigue/exercise intolerance**, (**AFib/AFlutter/ Chief Complaint**)

16(S) Clutching chest:
AFib/AFlutter can present with **chest pain** in elderly patients → demand ischemia from arrhythmia? or ischemia itself inducing the arrhythmia?)

17(S) Old dude:
AFib/AFlutter risk factors → **age (older) and sex (male) (AFib/AFlutter/ Demographics**)

18(S) Strained pressurized heater:
Factors that put strain on atria (↑size, ↑pressure, ↑inflammation) predispose to development of AFib/AFlutter → **hypertensive heart disease** and coronary heart disease (w/ hx of MI or heart failure) are the most important factors in developed countries **(AFib/AFlutter/ PMHx)**

19(S) Broken heart strings and floppy heart balloon:
Factors that put strain on atria (↑size, ↑pressure, ↑inflammation) predispose to development of AFib/AFlutter → hypertensive heart disease and **coronary heart disease (w/ hx of MI or heart failure)** are the most important factors in developed countries **(AFib/AFlutter/ PMHx)**

20(S) Faulty valve and “rheu-barb” pie:
Factors that put strain on atria (↑size, ↑pressure, ↑inflammation) predispose to development of AFib/AFlutter → valvular heart disease (regurg/stenosis), **valvular rheumatic heart disease** (↑prevalence in developing countries) **(AFib/AFlutter/ PMHx)**

21(S) Heart knife:
Factors that put strain on atria (↑size, ↑pressure, ↑inflammation) predispose to development of AFib/AFlutter → **cardiac surgery (esp CABG and valve replacement surgery) (AFib/AFlutter/ PMHx)**

22(S) Sleep mask + thrombotic nest in pulmonary tree:
Factors that put strain on atria (↑size, ↑pressure, ↑inflammation) predispose to development of AFib/AFlutter → **pulmonary disease (e.g. PE, OSA)** → right atrial strain (cor pulmonale) **(AFib/AFlutter/ PMHx)**

23(S) Obese belly + candy jar:
Factors that put strain on atria (↑size, ↑pressure, ↑inflammation) predispose to development of AFib/AFlutter → **obesity and diabetes** (multifactorial) (**AFib/AFlutter/ PMHx**)

24(S) Excessive alcohol:
Factors that put strain on atria (↑size, ↑pressure, ↑inflammation) predispose to development of AFib/AFlutter → **binge drinking (“holiday heart syndrome”) and chronic alcohol abuse (alcoholic [dilated] cardiomyopathy); (AFib/AFlutter/ PMHx)**

25(S) Thrombotic poops to head, abdomen, and extremities:
episodic or chronic AFib/AFlutter causes loss of coordinated atrial contraction/ stagnant blood → blood clot in atrium → may present with **thromboembolic events** (e.g. stroke/TIA, acute limb/visceral ischemia) (**AFib/AFlutter/ Chief Complaint**)

26) Rhythm record, rate metronome, and anti-clog:
in patients with an established history of AFib/Aflutter, assess home meds, past tx strategies, compliance → **Rhythm control?** (e.g. amiodarone, dofetilide, flecainide, propafenone, sotalol, dronedarone) **Rate control?** (e.g. beta blocker, verapamil, diltiazem, digoxin) **Anticoagulation?** (e.g. warfarin, NOAC)

27(O) BP cuff, failing heart balloon:
AFib/AFlutter can present with signs/symptoms of **acute/chronic heart failure → hypotension** (dizziness/lightheadedness/**AMS**), **periph edema, JVD, pulmonary edema, cardiac gallop**, exertional dyspnea/fatigue/exercise intolerance (**AFib/AFlutter/ Vitals/Physical Exam**)

28(O) ECG grid wallpaper:
Inspect ECG for features of AFib/AFlutter (rate, rhythm, pf/F wave morphology) → **Other SVTs? Signs of ischemia? Structural abnormalities (BBB, A/V enlargement)? (AFib/AFlutter/ ECG**)

29(O) T-bone dinner:
AFib/AFlutter can cause demand ischemia from arrhythmia, or ischemia itself can induce the arrhythmia → **monitor troponins** (if clinical scenario indicates, e.g. older patient with CAD, chest pain, ischemic changes on ECG, or cardiac risk factors) (**AFib/AFlutter/ Labs**)

30(O) “Lytes” beverage:
Order **metabolic panel + lytes** in order to assess electrolytes that participate in myocyte depolarization (Na, K, Ca, Mg); assess renal function (altered hemodynamics) (**AFib/AFlutter/ Labs**)

31(O) Tightening thyroid bowtie:
Hyperthyroidism is associated with AFib/AFlutter → consider checking **TSH** in select patients (especially those with new onset of AFib/AFlutter) (**AFib/AFlutter/ Labs**)

32(O) Echo bullhorn at cardio pool:
Order a **transthoracic echo (TTE)** as part of the routine w/u of AFib/AFlutter → assess for underlying heart disease (e.g. MI, valvular, pericardial, LVH, HF) that might be precipitant of arrhythmia; LV function also dictates tx strategy (**AFib/AFlutter/ TTE**)

33(O) Enlarged left atrial lady:
Order a transthoracic echo (TTE) as part of the routine w/u of AFib/AFlutter → assess for **LA dilation** which can precipitate arrhythmia or is caused by the arrhythmia (in which case, RA dilation is reversed after restoration of sinus rhythm) (**AFib/AFlutter/ TTE**)

34(O) Obscured thrombotic stick pile in upper left chamber:
transthoracic echo (TTE) is NOT sensitive (30-60%) for identifying a left atrial clot; TEE is more sensitive for detection of left atrial thrombus, but only used when affecting management decisions (e.g. cardioversion) (**AFib/AFlutter/ TTE**)==O== Three different P-shaped yogis on the MAT

35(O) Three different P-shaped yogis on the MAT:
Multifocal atrial tachycardia (MAT) involves **pacemaker activity in ≥3 areas of the atria** (possibly single focus with ≥3 conduction paths?) → **at least 3 distinct P wave morphologies** → impulses sent through AV node to ventricles (irregular narrow complex tachycardia) (**MAT/ ECG**)

36(O) More irregularly irregular static on screen:
Multifocal atrial tachycardia (MAT) involves pacemaker activity in ≥3 areas of the atria (possibly single focus with ≥3 conduction paths?) → at least 3 distinct P wave morphologies → impulses sent through AV node to ventricles (**irregular narrow complex tachycardia**) (**MAT/ ECG**)

37(O) Focused on single “P” yogi:
Focal atrial tachycardia (FocalAT) involves **pacemaker activity in a single focus of the atria → single P wave morphology (usually abnormal morphology, esp if focus is distant from SA node)** → impulses sent through AV node to ventricles (regular narrow complex tachycardia) (**FocalAT/ ECG**)

38(O) Regular notes:
Focal atrial tachycardia (FocalAT) involves pacemaker activity in a single focus of the atria → single P wave morphology (usually abnormal morphology, esp if focus is distant from SA node) → impulses sent through AV node to ventricles (**regular narrow complex tachycardia**) (**FocalAT/ ECG**)

39(O) ==P== “Just treat it” shirt:
Focal atrial tachycardia (FocalAT) → **assess/ treat underlying conditions** that can precipitate atrial arrhythmia (HF, cardiomyopathy, MI, infection, electrolytes) → NONsustained FocalAT is typically benign in adults (SUSTAINED Atach can lead to a tachycardia-mediated cardiomyopathy) → tx for recurrent symptomatic dz = BB, CBB, or ablation (**FocalAT/ ECG/ Plan**)

40(S) Blue bloater:
Multifocal atrial tachycardia (MAT) is particularly associated with **pulmonary disease/ acute pulmonary failure** (**especially COPD**, also pneumonia, PE) → also associated with heart dz (e.g.CAD, valvular, HTN) with HF → assess for other conditions that can precipitate atrial arrhythmia (electrolytes, drugs) (**MAT/ Chief Complaint/ PMHx**)

41(S) Lung stains and thrombotic nest in pulm tree:
Multifocal atrial tachycardia (MAT) → particularly associated with pulmonary disease/ acute pulmonary failure (especially COPD, also **pneumonia, PE**) → also associated with heart dz (e.g.CAD, valvular, HTN) with HF → assess for other conditions that can precipitate atrial arrhythmia (electrolytes, drugs) (**MAT/ Chief Complaint/ PMHx**)

42(S) ==O== Failing heart balloon and electrolyte beverage:
Multifocal atrial tachycardia (MAT) → particularly associated with pulmonary disease/ acute pulmonary failure (especially COPD, also pneumonia, PE) → also associated with **heart dz (e.g. CAD, valvular, HTN) with HF** → assess for other conditions that can precipitate atrial arrhythmia (**electrolytes,** drugs) (**MAT/ PMHx/ Medications/ Labs**)

43(S) Struggling to breath:
Multifocal atrial tachycardia (MAT) → particularly associated with pulmonary disease/ acute pulmonary failure (especially COPD, also pneumonia, PE) → patients typically **present with sx 2/2 underlying condition** rather than sx 2/2 arrhythmia (**MAT/ Chief Complaint**)

44(P) “Just treat it”:
MAT is usually self-limited and resolves with **tx of the underlying causal disease process** (e.g. pulmonary disease/ failure) (**MAT/ Plan**)

45(P) Electrolyte replenishment:
MAT is usually self-limited and resolves with tx of the underlying causal disease process (e.g. pulmonary disease/ failure, **treat electrolyte deficiencies**) (**MAT/ Plan**)

46(P) Calci-yum ice cream and muted beta bugle:
Patient with MAT persistent AND symptomatic → treat with **beta blockers/ calcium channel blockers** to slow ventricular rate → cardioversion or rhythm control agents are not usually pursued